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Arithmetic Inequalities

By the end of Friday, Jane had recorded a total of 4 + 2 = 6 hours of programs, and she had viewed between 1 + 2 = 3 hours and 2 + 3 = 5 hours of these programs. Therefore, the number of hours of recorded programs that Jane had not yet viewed by the end of Friday was between 6 − 5 = 1 and 6 − 3 = 3.

The correct answer is E.

•

A dance troupe has a total of 50 dancers split into 2 groups. The costumes worn by Group A cost $80 each, and those worn by Group B cost $90 each. If the total cost of all the costumes is $4,270, what is the total cost of the costumes worn by Group B?

A. $1,840

B. $2,070

C. $2,135

D. $2,160

E. $2,430

Algebra Applied Problems; First-Degree Equations

Let n be the number of dancers in Group B. Then the number of dancers in Group A is $50 - n$, and we are given that $(50 - n)(\$80) + (n)(\$90) = \$4,270$.

$$\begin{aligned} (50-n)(80)+90n &= 4,270 && \text{given} \\ (50-n)(8)+9n &= 427 && \text{divide both sides by 10} \\ 400-8n+9n &= 427 && \text{distributive law} \\ n &= 27 && \text{combine like terms and subtract 400 from each side} \end{aligned}$$

Therefore, the total cost of the costumes worn by Group B is 27($90) = $2,430.

The correct answer is E.

•

A doctor prescribed 18 cubic centimeters of a certain drug to a patient whose body weight was 120 pounds. If the typical dosage is 2 cubic centimeters per 15 pounds of body weight, by what percent was the prescribed dosage greater than the typical dosage?

A. 8%

B. 9%

C. 11%

D. 12.5%

E. 14.8%

Arithmetic Percents

If the typical dosage is 2 cubic centimeters per 15 pounds of body weight, then the typical dosage for a person who weighs 120 pounds is $2\left(\frac{120}{15}\right) = 2(8) = 16$ cubic centimeters. The prescribed dosage of 18 cubic centimeters is, therefore, $\left(\left(\frac{18-16}{16}\right) \times 100\right)\%$ or 12.5% greater than the typical dosage.

The correct answer is D.

•

The function f is defined by $f(x) = \sqrt{x} - 10$ for all positive numbers x. If $u = f(t)$ for some positive numbers t and u, what is t in terms of u?

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and u, what is t in terms of u?

A. $\sqrt{\sqrt{u}+10}$

B. $(\sqrt{u}+10)^2$

C. $\sqrt{u^2+10}$

D. $(u+10)^2$

E. $(u^2+10)^2$

Algebra Functions

Because $f(x)=\sqrt{x}-10$, it follows that $f(t)=\sqrt{t}-10$. The problem states that $u=f(t)$, so by substitution, $u=\sqrt{t}-10$. To express t in terms of u, isolate t on one side of the equation:

$$\begin{aligned} u &= \sqrt{t}-10 && \text{given} \\ u+10 &= \sqrt{t} && \text{add 10 to both sides} \\ (u+10)^2 &= t && \text{square both sides} \end{aligned}$$

Therefore, t in terms of u is $(u+10)^2$.

The correct answer is D.

•

If 35% of a number is decreased by 15, then the result is 25% of the original number. What is the original number?

A. 1.5

B. 9

C. 25

D. 150

E. 900

Arithmetic Percents

We need to find x such that $0.35x-15=0.25x$. Adding $15-0.25x$ to both sides of the equation gives us $0.1x=15$, so $x=150$.

The correct answer is D.

•

A flower arrangement consists of 30 roses, each of which is either white or red. If a rose is to be selected at random from the flower arrangement, the probability that the rose selected will be white is twice the probability that it will be red. How many white roses are in the flower arrangement?

A. 5

B. 10

C. 15

D. 20

E. 25

Algebra Applied Problems

Let W be the number of white roses and R be the number of red roses. From the given information, $W+R=30$ and $\frac{W}{30}=2\times\frac{R}{30}$, where $\frac{W}{30}$ is the probability that the rose selected will be white and $\frac{R}{30}$ is the probability that the rose selected will be red. Multiplying both sides of $\frac{W}{30}=2\times\frac{R}{30}$ by 30 gives $W=2R$. Substituting $W=2R$

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into $W + R = 30$ gives $2R + R = 30$, or $R = 10$. Therefore, $W = 2R = 2(10) = 20$.

The correct answer is D.

•

Sixty percent of the people who responded to a company's advertisement for a job opening were interviewed over the phone. Twenty percent of those interviewed over the phone, that is, 3 people, were asked to come to the company for a personal interview. How many people responded to the advertisement?

A. 16

B. 20

C. 25

D. 40

E. 64

Arithmetic Percents

Since 20% of the 60% who responded were the 3 people who were asked to come for an interview, it follows that 12% of those who responded were the 3 people who were asked to come for an interview. Therefore, the number of those who responded is 3 ÷ 0.12 =25.

The correct answer is C.

•

$(-3)^{-2} =$

A. – 9

B. – 6

C. $-\frac{1}{9}$

D. $\frac{1}{9}$

E. 9

Arithmetic Negative Exponents

$$(-3)^{-2} = \frac{1}{(-3)^2} = \frac{1}{9}$$

The correct answer is D.

•

Mark bought a set of 6 flowerpots of different sizes at a total cost of $8.25. Each pot cost $0.25 more than the next one below it in size. What was the cost, in dollars, of the largest pot?

A. $1.75

B. $1.85

C. $2.00

D. $2.15

E. $2.30

Algebra Applied Problems; First-Degree Equations

Let C be the cost of the largest pot in dollars. Then, in dollars, the total cost 8.25 is equal to $C + (C - 0.25) + (C - 0.50) + (C - 0.75) + (C - 1.00) + (C - 1.25)$, or $6C - 3.75$. Therefore, $6C - 3.75 = 8.25$, which when solved gives $C = 2$.

The correct answer is C.

•

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•

If r and s are positive integers and $r - s = 6$, which of the following has the greatest value?

A. $2r$

B. $2s$

C. $r + s$

D. $2r - s$

E. $2s - r$

Arithmetic Order

Since we can assume that only one of the answer choices has the greatest value, any two positive integers r and s such that $r - s = 6$ can be used to identify that answer choice. The table below shows the values of the answer choices for $r = 7$ and $s = 1$.

answer choice	value when $r = 7$ and $s = 1$
$2r$	14
$2s$	2
$r + s$	8
$2r - s$	13
$2s - r$	−5

From the table it follows that the first answer choice has the greatest value.

Alternatively, an algebraic approach can be used to identify the greatest value without making the assumption that only one of the answer choices always has the greatest value. First, using $s = r - 6$, express each of the answer choices in terms of r. For example, in terms of r the answer choice $2s - r$ is equal to $2(r - 6) - r = r - 12$. The table below shows the answer choices in terms of r.

answer choice	answer choice in terms of r
$2r$	$2r$
$2s$	$2r - 12$
$r + s$	$2r - 6$
$2r - s$	$r + 6$
$2s - r$	$r - 12$

From the second column of the table, it follows that, for each possible value of r, the first answer choice $2r$ is the greatest of the first three answer choices and the fourth answer choice $2r - s = r + 6$ is the greater of the remaining two answer choices. Therefore, the greater of the first and fourth answer choices, $2r$ and $r + 6$, will be the greatest of the answer choices. To determine which of $2r$ and $r + 6$ is greater, note that since s is a positive integer, we have $s + 6 > 6$. Now use $r = s + 6$ to get $r > 6$, from which it follows that $r + r > r + 6$, or $2r > r + 6$.

The correct answer is A.

•

An auction house charges a commission of 15 percent on the first \$50,000 of the sale price of an item, plus 10 percent on the amount of the sale price in excess of \$50,000. What was the sale price of a painting for which the auction house charged a total commission of \$24,000?

A. \$115,000

B. \$160,000

C. \$215,000

D. \$240,000

E. \$365,000

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Arithmetic Applied Problems; *Percents*

Since the commission on \$50,000 is (0.15)(\$50,000) = \$7,500 and the total commission of \$24,000 is \$16,500 greater than \$7,500, it follows that \$16,500 of the total commission arises from the sale price in excess of \$50,000. Therefore, 10% of the sale price in excess of \$50,000 is equal to \$16,500, and hence the sale price in excess of \$50,000 is \$165,000 (because 10% of \$165,000 is \$16,500). Thus, the sale price is \$50,000 + \$165,000 = \$215,000.

Alternatively, let P be the sale price in dollars. Then the given information can be expressed as a first-degree equation which can be solved for the value of P.

$$24{,}000 = 0.15(50{,}000) + 0.10(P - 50{,}000)$$
$$24{,}000 = 7,500 + 0.10P - 5{,}000$$
$$21,500 = 0.10P$$
$$P = 215{,}000$$

The correct answer is C.

•

At a certain college there are twice as many English majors as history majors and three times as many English majors as mathematics majors. What is the ratio of the number of history majors to the number of mathematics majors?

A. 6 to 1

B. 3 to 2

C. 2 to 3

D. 1 to 5

E. 1 to 6

Algebra Ratios

Let E, H, and M denote, respectively, the number of English, history, and mathematics majors. From the given information, it follows that $E = 2H$ and $E = 3M$, or equivalently, $H = \frac{1}{2}E$ and $M = \frac{1}{3}E$. Therefore,

$\frac{H}{M} = \frac{\frac{1}{2}}{\cancel{E}} = \frac{3}{2}$, and hence the ratio of the number of history majors to the number of mathematics majors is 3 to 2.

The correct answer is B.

•

A certain method for calculating pediatric drug dosages states that if d is the adult dosage, in milligrams, and y is the child's age, in years, then the child's dosage, in milligrams, is $d\left(\frac{y+1}{24}\right)$. If the adult dosage for a particular drug is 800 milligrams, what is an 8-year-old child's dosage, in milligrams, according to this method?

A. 200

B. 225

C. 250

D. 275

E. 300

Algebra Formulas

From the given information, we know that $d = 800$ and $y = 8$. Therefore, the dosage, in milligrams, for the 8-year-old child is

$d\left(\frac{y+1}{24}\right) = 800\left(\frac{8+1}{24}\right) = 800 \times \frac{3}{8} = 300.$

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$a\left(\frac{}{24}\right) = 800\left(\frac{}{24}\right) = 800 \times \frac{}{8} = 300.$

The correct answer is E.

Questions 97 to 176 — Difficulty: Medium

97. The sides of each of two plastic cubes are numbered 1 through 6, and each number is equally likely to appear face up after either cube is rolled. What is the probability that after the two cubes are rolled, the sum of the two numbers appearing face up will be greater than 9?

A. $\frac{1}{6}$

B. $\frac{1}{5}$

C. $\frac{1}{4}$

D. $\frac{5}{18}$

E. $\frac{1}{3}$

Arithmetic Statistics

For either cube, any 1 of the 6 numbers is equally likely to appear face up after rolling the cube. Thus, $6 \times 6 = 36$ equally likely outcomes of rolling both cubes are possible. Of these 36, only 6 outcomes have sums greater than 9. These are {4, 6}, {5, 5}, {5, 6}, {6, 4}, {6, 5}, and {6, 6}, where the first number in each set is face up on the first cube and the second is face up on the second cube. Since each outcome is equally probable, the probability of the sum of the numbers facing up being greater than 9 is $\frac{6}{36} = \frac{1}{6}$.

The correct answer is A.

98. On Saturday morning, Malachi will begin a camping vacation, and he will return home at the end of the first day on which it rains. If on the first three days of the vacation the probability of rain on each day is 0.2, what is the probability that Malachi will return home at the end of the day on the following Monday?

A. 0.008

B. 0.128

C. 0.488

D. 0.512

E. 0.640

Algebra Statistics

The first three days of Malachi's vacation are Saturday, Sunday, and Monday. On each of these days, if Malachi is still camping, the probability that it will rain and thus he'll return home that day is 0.2, which means the probability that he won't return home that day is $1 - 0.2 = 0.8$. So, the probability that he'll still be camping Sunday is 0.8, and the probability that he'll still be camping Monday is $0.8 \times 0.8 = 0.64$. Therefore, the probability that he'll return home Monday is $0.64 \times 0.2 = 0.128$.

The correct answer is B.

99. According to a survey of 200 people, 60 enjoy skiing and 80 enjoy skating. If the number of people who enjoy neither skiing nor skating is 2 times the number of people who enjoy both skiing and skating, how many people surveyed enjoy neither skiing nor skating?

A. 20

B. 40

C. 50

D. 80

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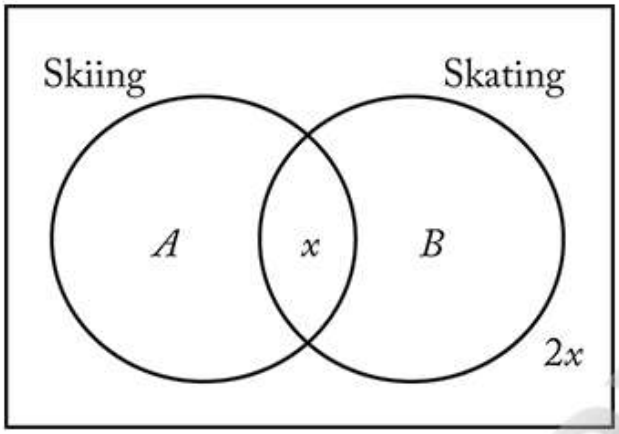
E. 120

Algebra Sets

Determine the number of people who enjoy neither skiing nor skating.



Consider the Venn diagram on the previous page, where A represents the number of people who enjoy skiing only, B represents the number of people who enjoy skating only, and x is the number of people who enjoy both skiing and skating. As stated, the number of people who enjoy neither is twice the number who enjoy both, so $2x$ represents the number of people who enjoy neither skiing nor skating. From the given information, it follows that $60 = A + x$, $80 = B + x$, and $200 = A + x + B + 2x = (A + x) + (B + x) + x$. By substitution, $200 = 60 + 80 + x$, so $x = 60$ and $2x = 120$.

The correct answer is E.

$$m \oplus p = n$$
$$n \oplus r = m$$
$$n \oplus q = q$$
$$p \oplus q = p$$
$$q \oplus p = r$$

100. If the relations shown hold for the operation $\oplus$ and the numbers m, n, p, q, and r, then $[(m \oplus p) \oplus q] \oplus p =$

A. m

B. n

C. p

D. q

E. r

Algebra Formulas

$$[(m \oplus p) \oplus q] \oplus p \text{ given}$$
$$[n \oplus q] \oplus p \; m \oplus p = n$$
$$q \oplus p \; n \oplus q = q$$
$$r \; q \oplus p = r$$

The correct answer is E.

101. To rent a tractor, it costs a total of x dollars for the first 24 hours, plus y dollars per hour for each hour in excess of 24 hours. Which of the following represents the cost, in dollars, to rent a tractor for 36 hours?

A. $x + 12y$

B. $x + 36y$

C. $12x + y$

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C. $12x + y$

D. $24x + 12y$

E. $24x + 36y$

Algebra Algebraic Expressions

Determine the cost to rent a tractor for 36 hours if the cost is x dollars for the first 24 hours and y dollars for each hour in excess of 24 hours.

The cost will be x dollars for the first 24 hours plus y dollars for each of 36 − 24 = 12 hours. Thus, the total cost of renting the tractor for 36 hours is $(x + 12y)$ dollars.

The correct answer is A.

102. If the mass of 1 cubic centimeter of a certain substance is 7.3 grams, what is the mass, in kilograms, of 1 cubic meter of this substance? (1 cubic meter = 1,000,000 cubic centimeters; 1 kilogram = 1,000 grams)

A. 0.0073

B. 0.73

C. 7.3

D. 7,300

E. 7,300,000

Arithmetic Measurement Conversion

Determine the mass, in kilograms, of 1 cubic meter of a substance if the mass of 1 cubic centimeter of this substance is 7.3 grams.

1 cubic meter is 1,000,000 cubic centimeters and each cubic centimeter has a mass of 7.3 grams, so 1 cubic meter has a mass of 7.3(1,000,000) = 7,300,000 grams. Then, since 1 kilogram = 1,000 grams,

$$7,300{,}000 \text{ grams} = \frac{7,300{,}000}{1{,}000} = 7,300 \text{ kilograms}.$$

The correct answer is D.

103. If $z \neq 0$ and $z + \frac{1-2z^2}{z} = \frac{w}{z}$, then $w =$

A. $z + 1$

B. $z^2 + 1$

C. $-z^2 + 1$

D. $-z^2 + z + 1$

E. $-2z^2 + 1$

Algebra Simplifying Algebraic Expressions

Multiplying both sides of $\frac{w}{z} = z + \frac{1-2z^2}{z}$ by z gives $w = z\left(z + \frac{1-2z^2}{z}\right) = z^2 + (1-2z^2) = 1-z^2$.

The correct answer is C.

104. For all real numbers a, b, c, d, e, and f, the operation Θ is defined by the equation $(a, b, c)\Theta(d, e, f) = ad + be + cf$. What is the value of $\left(1, -2, 3\right)\Theta\left(1, -\frac{1}{2}, \frac{1}{3}\right)$?

A. −1

B. $\frac{5}{6}$

C. 1

D. 5

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D. $\frac{5}{2}$

E. 3

Algebra Formulas

By definition, $\left(1,-2,3\right)\Theta = \left(1,-\frac{1}{2},\frac{1}{3}\right) = (1)(1) + (-2)\left(-\frac{1}{2}\right) + (3)\left(\frac{1}{3}\right) = 1 + 1 + 1 = 3.$

The correct answer is E.

ABSENCES FROM A CERTAIN MATH CLASS DURING THE SPRING SEMESTER	
Number of Absences	Number of Students
0	4
1	3
2	10
3	3
4	5
5 or more	3

105. The table above shows the distribution of the number of absences from a certain Math class during the spring semester. For those students who had at least 1 absence, what was the median number of absences?

A. 1.5

B. 2

C. 2.5

D. 3

E. 3.5

Arithmetic Statistics

The table shows that the total number of students who each had at least 1 absence was 3 + 10 + 3 + 5 + 3 = 24. To find the median number of absences for these 24 students, we list the students' numbers of absences from least to greatest, giving us a list of 24 numbers (one per student): three 1s, ten 2s, three 3s, five 4s, and three numbers that are each at least 5. Since 24 is even, the median is the mean of the two middle numbers in the list —the twelfth number and the thirteenth number. These numbers are both 2s. So, 2 is the median.

The correct answer is B.

106. If $\frac{x}{y} = \frac{c}{d}$ and $\frac{d}{c} = \frac{b}{a}$, which of the following must be true?

I. $\frac{y}{x} = \frac{b}{a}$

II. $\frac{x}{a} = \frac{y}{b}$

III. $\frac{y}{a} = \frac{x}{b}$

A. I only

B. II only

C. I and II only

D. I and III only

E. I, II, and III

Algebra Ratio and Proportion

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Equation I is true:

$\frac{x}{y} = \frac{c}{d}$ given

$\frac{y}{x} = \frac{d}{c}$ take reciprocals

$\frac{d}{c} = \frac{b}{a}$ given

$\frac{y}{x} = \frac{b}{a}$ use last two equations

Equation II is true:

$\frac{y}{x} = \frac{b}{a}$ Equation I (shown true)

$y = \frac{bx}{a}$ multiply both sides by x

$\frac{y}{b} = \frac{x}{a}$ divide both sides by b

Equation III is false, since otherwise it would follow that:

$y = \frac{bx}{a}$ from earlier

$\frac{y}{a} = \frac{bx}{a^2}$ divide both sides by a

$\frac{x}{b} = \frac{bx}{a^2}$ use Equation III (assumed true)

$x = \frac{b^2x}{a^2}$ multiply both sides by b

From this it follows that Equation III will hold only if $\frac{b^2}{a^2} = 1$, which can be false. For example, if $x = a = c = 1$ and $y = b = d = 2$ (a choice of values for which $\frac{x}{y} = \frac{c}{d}$ and $\frac{d}{c} = \frac{b}{a}$ are true), then $\frac{b^2}{a^2} \neq 1$ and Equation III is $\frac{2}{1} = \frac{1}{2}$, which is false.

The correct answer is C.

107. If $\lceil x \rceil$ denotes the least integer greater than or equal to x and $\left\lceil \frac{x}{2} \right\rceil = 0$, which of the following could be the value of x?

A. –2

B. $-\frac{3}{2}$

C. $\frac{1}{2}$

D. 1

E. 2

Arithmetic Formulas

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Since the least integer greater than or equal to $\frac{x}{2}$ is 0, then $-1 < \frac{x}{2} \leq 0$. Thus, $-2 < x \leq 0$. Among the 5 options, $-\frac{3}{2}$ is the only number greater than −2 and less than or equal to 0, and therefore the only one that could be the value of x.

The correct answer is B.

108. If $a(a+2) = 24$ and $b(b+2) = 24$, where $a \neq b$, then $a + b =$

A. –48

B. –2

C. 2

D. 46

E. 48

Algebra Second-Degree Equations

$a(a+2) = 24$	given
$a^2 + 2a = 24$	use distributive property
$a^2 + 2a - 24 = 0$	subtract 24 from both sides
$(a+6)(a-4) = 0$	factor

So, $a + 6 = 0$, which means that $a = -6$, or $a - 4 = 0$, which means $a = 4$. The equation with the variable b has the same solutions, and so $b = -6$ or $b = 4$.

Since $a \neq b$, then $a = -6$ and $b = 4$, which means $a + b = -6 + 4 = -2$, or $a = 4$ and $b = -6$, which means that $a + b = 4 + (-6) = -2$

The correct answer is B.

109. In a recent election, Ms. Robbins received 8,000 votes cast by independent voters, that is, voters not registered with a specific political party. She also received 10 percent of the votes cast by those voters registered with a political party. If N is the total number of votes cast in the election and 40 percent of the votes cast were cast by independent voters, which of the following represents the number of votes that Ms. Robbins received?

A. $0.06N + 3,200$

B. $0.1N + 7,200$

C. $0.4N + 7,200$

D. $0.1N + 8,000$

E. $0.06N + 8,000$

Algebra Percents

If N represents the total number of votes cast and 40% of the votes cast were cast by independent voters, then 60% of the votes cast, or $0.6N$ votes, were cast by voters registered with a political party. Ms. Robbins received 10% of these, and so Ms. Robbins received $(0.10)(0.6N) = 0.06N$ votes cast by voters registered with a political party. Thus, Ms. Robbins received $0.06N$ votes cast by voters registered with a political party and 8,000 votes cast by independent voters, so she received $0.06N + 8,000$ votes in all.

The correct answer is E.

110. The profit P, in dollars, for any given month at a certain company is defined by $P = I - C$, where I represents total income, in dollars, and C represents total costs, in dollars, for the month. For each of the first 4 months of the year, $C = I + 32,000$; and for each of the next 3 months, $I = C + 36,000$. If $I = C + 10,000$ for each of the 5 remaining months of the year, what was the company's total profit for the 12-month year?

A. $10,000

B. $20,000

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B. $30,000

C. $40,000

D. $50,000

E. $70,000

Arithmetic Applied Problems

For each of the first 4 months, total costs exceeded total income by $32,000, and hence for each of 4 months total profit was −$32,000. Also, for each of the next 3 months, total income exceeded total costs by $36,000, and hence for each of 3 months total profit was $36,000. Finally, for each of the remaining 5 months, total income exceeded total costs by $10,000, and hence for each of 5 months total profit was $10,000. Therefore, the total profit for the 12-month year was 4(–$32,000) + 3($36,000) + 5($10,000) = $30,000.

Alternatively, the given equations can be easily manipulated to obtain the values of $I - C$ for each month, as shown below.

Given Equation	Value of $I - C$	Number of Months	Net Profit
$C = I + 32{,}000$	$I - C = -32{,}000$	4	−$128,000
$I = C + 36{,}000$	$I - C = 36{,}000$	3	$108,000
$I = C + 10{,}000$	$I - C = 10{,}000$	5	$50,000

Adding the values in the right-most column gives the total profit for the 12-month year: −$128,000 + $108,000 + $50,000 = $30,000.

The correct answer is B.

111. A manufacturer makes and sells 2 products, P and Q. The revenue from the sale of each unit of P is $20.00 and the revenue from the sale of each unit of Q is $17.00. Last year the manufacturer sold twice as many units of Q as P. What was the manufacturer's average (arithmetic mean) revenue per unit sold of these 2 products last year?

A. $28.50

B. $27.00

C. $19.00

D. $18.50

E. $18.00

Arithmetic Statistics

Let x represent the number of units of Product P the manufacturer sold last year. Then $2x$ represents the number of units of Product Q the manufacturer sold, and $x + 2x = 3x$ represents the total number of units of Products P and Q the manufacturer sold. The total revenue from the sale of Products P and Q was $\$(20x) + \$(17(2x)) = \$(54x)$, so the average revenue per unit sold was $\frac{\$(54x)}{3x} = \$18$.

The correct answer is E.

112. A worker carries jugs of liquid soap from a production line to a packing area, carrying 4 jugs per trip. If the jugs are packed into cartons that hold 7 jugs each, how many jugs are needed to fill the last partially filled carton after the worker has made 17 trips?

A. 1

B. 2

C. 4

D. 5

E. 6

Arithmetic Remainders

Carrying 4 jugs per trip, the worker carries a total of 4(17) = 68 jugs in 17 trips. At 7 jugs per carton, these jugs

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will completely fill 9 cartons with 5 jugs left over since (9)(7) + 5 = 68. To fill the 10th carton, 7 – 5 = 2 jugs are needed.

The correct answer is B.

113. Last year a state senate consisting of only Republican and Democrat members had 20 more Republican members than Democrat members. This year the senate has the same number of members as last year, but it has 2 fewer Republican members than last year. If this year the number of Republican members is $\frac{2}{3}$ the number of senate members, how many members does the senate have this year?

A. 33

B. 36

C. 42

D. 45

E. 48

Algebra First-Degree Equations

Let D be the number of Democrat members last year. Then $D + 20$ is the number of Republican members last year and $(D + 20) - 2 = D + 18$ is the number of Republican members this year. Since the total number of members this year is the same as last year, it follows that there are 2 more Democrat members this year than last year, and so the number of Democrat members this year is $D + 2$ and the total number of members this year is $2D + 20$. Using the fact that this year the number of Republican members is $\frac{2}{3}$ the total number of members, the value of D can be found.

$$D + 18 = \frac{2}{3}\left(2D + 20\right) \quad \text{given}$$

$$3(D + 18) = 2(2D + 20) \quad \text{multiply both sides by 3}$$

$$3D + 54 = 4D + 40 \quad \text{distributive law}$$

$$D = 14 \quad \text{subtract both } 3D \text{ and 40 from both sides}$$

Therefore, the total number of members this year is $2D + 20 = 2(14) + 20 = 48$.

The correct answer is E.

114. Sam has $800 in his account. He will deposit $1 in his account one week from now, $2 two weeks from now, and each week thereafter he will deposit an amount that is $1 greater than the amount that he deposited one week before. If there are no other transactions, how much money will Sam have in his account 50 weeks from now?

A. $850

B. $1,200

C. $1,675

D. $2,075

E. $3,350

Arithmetic Series and Sequences

Determine the amount Sam will have in his account after 50 weeks if the account starts with $800 and Sam deposits $1 more each week than he deposited the week before.

Sam's deposits over the 50-week period will be 1 +2 + 3 + … + 48 + 49 + 50 = (1 + 50) + (2 + 49) + (3 + 48) + … + (25 + 26) = (25)(51) = $1,275. Thus, after 50 weeks Sam will have $800 + $1,275 = $2,075.

The correct answer is D.

115. A certain state's milk production was 980 million pounds in 2007 and 2.7 billion pounds in 2014. Approximately how many more million gallons of milk did the state produce in 2014 than in 2007? (1 billion = 10^9 and 1 gallon

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how many more million gallons of milk did the state produce in 2014 than in 2007? (1 billion = 10^9 and 1 gallon = 8.6 pounds.)

A. 100

B. 200

C. 1,700

D. 8,200

E. 14,800

Arithmetic Measurement Conversion

Using 2.7 billion = 2,700 million, the amount of increase in millions of gallons can be calculated as follows.

The amount of increase is:

(2,700 – 980) million pounds = 1,720 million pounds

To convert pounds into gallons:

$$\left(1,720 \text{ million } \cancel{\text{pounds}}\right)\left(\frac{1 \text{ gallon}}{8.6 \cancel{\text{pounds}}}\right) = 200 \text{ million gallons}$$

The correct answer is B.

116. Working simultaneously and independently at an identical constant rate, 4 machines of a certain type can produce a total of x units of Product P in 6 days. How many of these machines, working simultaneously and independently at this constant rate, can produce a total of $3x$ units of Product P in 4 days?

A. 24

B. 18

C. 16

D. 12

E. 8

Algebra Applied Problems

Define a *machine day* as 1 machine working for 1 day. Then 4 machines each working 6 days is equivalent to (4)(6) = 24 machine days. Thus, x units of Product P were produced in 24 machine days, and $3x$ units of Product P will require (3)(24) = 72 machine days, which is equivalent to $\frac{72}{4} = 18$ machines working independently and simultaneously for 4 days.

The correct answer is B.

117. The symbol Δ denotes one of the four arithmetic operations: addition, subtraction, multiplication, or division. If 6 Δ 3 ≤ 3, which of the following must be true?

I. 2 Δ 2 = 0

II. 2 Δ 2 = 1

III. 4 Δ 2 = 2

A. I only

B. II only

C. III only

D. I and II only

E. I, II, and III

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filippo.fiz@uptoten.it